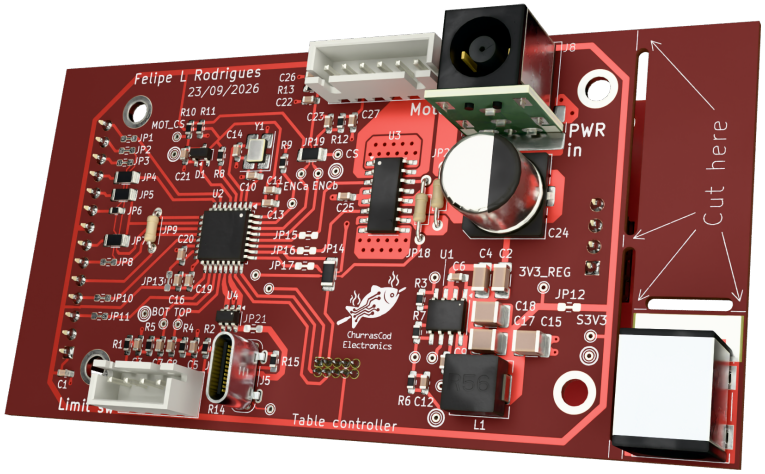
Desk controller assembly drawing

Top view



Bottom view



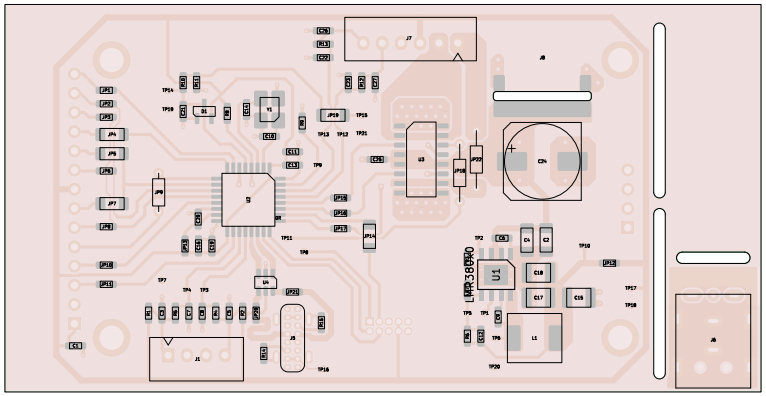
[Interactive BOM]

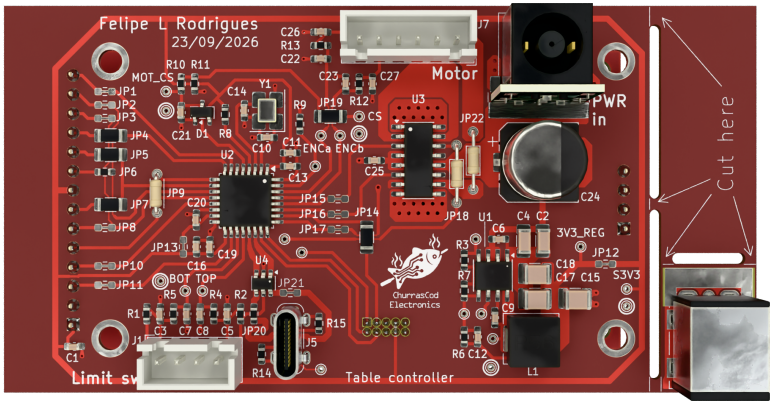
ChurrasCod Electronics

Desk controller	Sheet: Assembly overview	
	File: Desk_controller.kicad_pcb	
Notes	Rev: 1.0.1	Drwn: FLR
	Mat: Concept	Rev'd: FLR
	Date: 06/10/2026	Rls'd: FLR
	Dwg: NA	Size: A4
	Generated with KiCad	Page: 1/7

Top assembly (Scale 1:1)

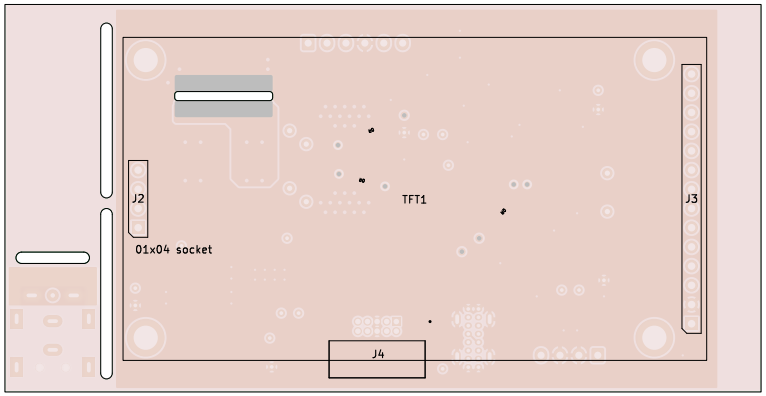
Type	Fitted	No fit
SMD	59	0
THT	6	0
Total	65	0





ChurrasCod Electronics	**Desk controller**	Sheet: Top assembly (Scale 1:1)
	Notes	File: Desk_controller.kicad_pcb
		Rev: 1.0.1 / Drwn: FLR
		Mat: Concept / Rev'd: FLR
		Date: 06/10/2026 / Rls'd: FLR
		Dwg: NA / Size: A4
		Generated with KiCad / Page: 2/7

Bottom assembly (Scale 1:1)





Type	Fitted	Not fitted
SMD	3	0
THT	4	0
Total	7	0

ChurrasCod Electronics

Desk controller

Notes

Sheet: Bottom assembly (Scale 1:1)
File: Desk_controller.kicad_pcb

Rev: 1.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 06/10/2026	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 3/7

Top BOM - fitted (1 of 3)

References	Qty	Value	Footprint	MPN	Manufacturer
C1 C6 C9 C13 C19 C20 C25	7	100n	C_0603_1608Metric_Pad1.08x0.95mm_HandSolder	AC0603KRX7R9BB104	Yageo
C3 C5 C26 C27	4	10n	C_0603_1608Metric_Pad1.08x0.95mm_HandSolder	AC0603KRX7R9BB103	Yageo
C7 C8 C21 C22 C23	5	1n	C_0603_1608Metric_Pad1.08x0.95mm_HandSolder	AC0603KRX7R9BB102	Yageo
C11 C16	2	1u	C_0603_1608Metric_Pad1.08x0.95mm_HandSolder	AC0603KRX7R7BB105	Yageo
C24	1	220u	CP_Elec_10x10	EEE-FK1H221P	Panasonic
C10 C14	2	22p	C_0603_1608Metric_Pad1.08x0.95mm_HandSolder	AC0603JRNPO9BN220	Yageo
C15 C17 C18	3	22u	C_1210_3225Metric_Pad1.33x2.70mm_HandSolder	MCAST32MSB7226KPNA01	TAIYO YUDEN
C12	1	33p	C_0603_1608Metric_Pad1.08x0.95mm_HandSolder	AC0603JRNPO9BN330	Yageo
C2 C4	2	4.7u	C_1206_3216Metric_Pad1.33x1.80mm_HandSolder	C1206C475K5RACAUTO	KEMET
D1	1	BAV99	SOT-23	BAV99Q-7-F	Diodes Inc
J1	1	B4B XH A	JST_XH_B4B-XH-A_1x04_P2.50mm_Vertical	B4B XH A	JST

ChurrasCod Electronics

Desk controller

Notes

Sheet: Top BOM - fitted (1 of 3)
File: Desk_controller.kicad_pcb
Rev: 1.0.1 | Drwn: FLR
Mat: Concept | Rev'd: FLR
Date: 06/10/2026 | Rls'd: FLR
Dwg: NA | Size: A4
Generated with KiCad | Page: 4/7

Top BOM - fitted (2 of 3)

References	Qty	Value	Footprint	MPN	Manufacturer
J7	1	B6B XH A	JST_XH_B6B-XH-A_1x06_P2.50mm_Vertical	B6B XH A	JST
J8	1	DC barrel assy	Dell_barrel_assy	NA	NA
J6	1	Dell DC jack	Dell DC jack	Barrel jack	Dell
J5	1	USB_C_Receptacle_USB2.0_16P	USB_C_socket_16P_vertical	TFKH5-16L4CC-65	ChengHaoRan
JP4 JP5 JP7 JP14 JP19	5	0R	R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	Hand cut wire	NA
JP18 JP22	2	0R	R_Axial_DIN0204_L3.6mm_D1.6mm_P7.62mm_Horizontal	Hand cut wire	NA
JP6 JP20	2	0R	R_0603_1608Metric_Pad0.98x0.95mm_HandSolder	RC0603JR-130RP	Yageo
JP26	1	0R	Wire_6.853x8.890mm	Hand cut wire	NA
JP9	1	0R	R_Axial_DIN0204_L3.6mm_D1.6mm_P5.08mm_Horizontal	Hand cut wire	NA
L1	1	10u	L_Bourns_SRP6540	SRP6540-100M	Bourns
R1 R2 R7 R8 R9	5	10k	R_0603_1608Metric_Pad0.98x0.95mm_HandSolder	AC0603FR-0710KL	YAGEO
R6	1	23.2k	R_0603_1608Metric_Pad0.98x0.95mm_HandSolder	AC0603FR-0723K2L	YAGEO

ChurrasCod Electronics

Desk controller

Notes

Sheet: Top BOM - fitted (2 of 3)
File: Desk_controller.kicad_pcb
Rev: 1.0.1 | Drwn: FLR
Mat: Concept | Rev'd: FLR
Date: 06/10/2026 | Rls'd: FLR
Dwg: NA | Size: A4
Generated with KiCad | Page: 5/7

Top BOM - fitted (3 of 3)

References	Qty	Value	Footprint	MPN	Manufacturer
R4 R5 R10 R12 R13	5	2k2	R_0603_1608Metric_Pad0.98x0.95mm_HandSolder	AC0603FR-072K2L	YAGEO
R14 R15	2	5k1	R_0603_1608Metric_Pad0.98x0.95mm_HandSolder	AC0603FR-075K1L	YAGEO
R3	1	64.9k	R_0603_1608Metric_Pad0.98x0.95mm_HandSolder	AC0603FR-0764K9L	YAGEO
R11	1	820	R_0603_1608Metric_Pad0.98x0.95mm_HandSolder	AC0603FR-07820RL	YAGEO
U1	1	LMR380x0	Texas_HSOP-8-1EP_3.9x4.9mm_P1.27mm_ThermalVias_w_step	LMR38020FDDAR	Texas instruments
U2	1	STM32G431K8T	LQFP-32_7x7mm_P0.8mm	STM32G431K8T6	ST
U4	1	USBLC6-2SC6	SOT-23-6	USBLC6-2SC6	ST
U3	1	VNH7070	JEITA_SOIC-16_3.9x9.9mm_P1.27mm	VNH7070	ST
Y1	1	20 MHz	Crystal_SMD_3225-4Pin_3.2x2.5mm_HandSoldering	CX3225SB20000H0FLJCC	KYOCERA AVX

ChurrasCod Electronics

Desk controller

Notes

Sheet: Top BOM - fitted (3 of 3)
File: Desk_controller.kicad_pcb
Rev: 1.0.1 | Drwn: FLR
Mat: Concept | Rev'd: FLR
Date: 06/10/2026 | Rls'd: FLR
Dwg: NA | Size: A4
Generated with KiCad | Page: 6/7

Bottom BOM - fitted (1 of 1)

References	Qty	Value	Footprint	MPN	Manufacturer
J2	1	01x04 socket	PinSocket_1x04_P2.54mm_Vertical	Generic	Generic
J3	1	01x14 socket	PinSocket_1x14_P2.54mm_Vertical	Generic	Generic
J4	1	HX JN1.27-2x5 WZ H4.9	JN1.27-2x5 WZ H4.9	HX JN1.27-2x5 WZ H4.9	Hanxia
JP23	1	0R	Wire_9.603mm	Hand cut wire	NA
JP24	1	0R	Wire_6.310mm	Hand cut wire	NA
JP25	1	0R	Wire_9.532mm	Hand cut wire	NA
TFT1	1	TFT_2.4in_320x240	Display_240x320_ILI9341_2.4in_no_pads	ILI9341_320x240	TENSTAR ROBOT

ChurrasCod Electronics

Desk controller

Notes

Sheet: Bottom BOM - fitted (1 of 1)
File: Desk_controller.kicad_pcb
Rev: 1.0.1 | Drwn: FLR
Mat: Concept | Rev'd: FLR
Date: 06/10/2026 | Rls'd: FLR
Dwg: NA | Size: A4
Generated with KiCad | Page: 7/7